

Subject Description Form

Subject Code	EEE531
Subject Title	Spoken Language Technologies
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	No
Objectives	This subject aims to provide a technology-centric and application-oriented understanding of state-of-the-art spoken language technologies that enable machines to interpret, understand and respond to human language. It equips students with the theoretical foundations, core algorithms, and practical considerations behind modern spoken language technologies, with a strong emphasis on recent advances in deep learning and neural networks that have revolutionized this field. Topics include speech analytics, spoken language recognition and synthesis, large language models, spoken language interfaces, speech privacy and security, and ethical considerations. The course is designed to keep students up to date with the latest developments and trends in spoken language technologies and what they can deliver in real-world industrial and commercial contexts. The course also features a hands-on <i>Speech AI Start-up Project</i>, allowing students to explore innovative applications and prototype ideas. In addition, invited lectures from leading researchers and industry practitioners provide real-world perspectives and exposure to cutting-edge developments.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand the fundamental principles and components of spoken language technologies, products, and services. - Analyze and evaluate state-of-the-art spoken language technologies and their practical capabilities. - Gain hands-on experience through lab sessions and projects, while critically assessing the limitations and challenges of current technologies. - Apply advanced spoken language technologies to design and develop applications and services.
Subject Synopsis/ Indicative Syllabus	Spoken Language Structure and Models - The human voice - Characteristics of speech - Pitch models Speech Perception and Quality Measures - Auditory perception - Psychoacoustics - Speech Quality Measures Modeling and Synthesis Tools - Short-time analysis and synthesis - Vector quantization - Linear predictive coding

	3.4 Vocoder 4. Speech Privacy and Security 4.1 Data privacy in speech communication 4.2 Speech synthesis 4.3 Speaker recognition and anonymization 4.4 Speech anti-spoofing 5. Spoken Language Generation 5.1 Language models (LMs) 5.2 Large language models (LLMs) 5.3 Instruction-tuned LLM and ChatGPT 6. Spoken Language Interface 6.1 Sequence-to-sequence model 6.2 Spoken dialogue systems 6.3 Dialogue system design																																								
Teaching/Learning Methodology	The theories and applications of spoken language technologies will be covered through lectures, providing students with both conceptual understanding and practical insights. Lab sessions will reinforce these concepts and offer hands-on experience with relevant tools and techniques. Additionally, students will work in groups to complete a project on selected topics – such as developing a start-up proposal – allowing them to apply their knowledge in a collaborative, real-world context. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Intended Subject Learning Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Lecture</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Tutorial</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>3. Laboratory</td><td></td><td></td><td>✓</td><td>✓</td></tr><tr><td>4. Project</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	Teaching/Learning Methodology	Intended Subject Learning Outcomes				a	b	c	d	1. Lecture	✓	✓	✓	✓	2. Tutorial	✓				3. Laboratory			✓	✓	4. Project	✓	✓	✓	✓											
Teaching/Learning Methodology	Intended Subject Learning Outcomes																																								
	a	b	c	d																																					
1. Lecture	✓	✓	✓	✓																																					
2. Tutorial	✓																																								
3. Laboratory			✓	✓																																					
4. Project	✓	✓	✓	✓																																					
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Laboratory reports</td><td>30%</td><td>✓</td><td></td><td>✓</td><td></td></tr><tr><td>2. Quiz</td><td>10%</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>3. Project</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>4. Examination</td><td>30%</td><td>✓</td><td>✓</td><td></td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td></td><td></td><td></td><td></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: 1. Lab Report: For each lab session, students will need to understand the fundamental concepts [Outcome (a)] before they can complete the lab exercises and write a report. Because the lab sessions involve the application of spoken language technologies [Outcome (c)], students’ ability to apply these technologies should be reflected in their reports. 2. Quiz: Quizzes will be given to check students understanding on the fundamental concepts. 3. Project: Students will need to conduct group projects on various spoken language technologies (e.g., Spoken Language Technology Start-Up proposal, pitching, storyboard), find out the limitations of these	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				a	b	c	d	1. Laboratory reports	30%	✓		✓		2. Quiz	10%	✓				3. Project	30%	✓	✓	✓	✓	4. Examination	30%	✓	✓		✓	Total	100%				
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed																																					
		a	b	c	d																																				
1. Laboratory reports	30%	✓		✓																																					
2. Quiz	10%	✓																																							
3. Project	30%	✓	✓	✓	✓																																				
4. Examination	30%	✓	✓		✓																																				
Total	100%																																								

	technologies [Outcome (d)], and determine what the current technologies can offer [Outcome (b)]. 4. Exam: Students will need to answer questions about the fundamental concepts [Outcome (a)] of various speech technologies and their applications [Outcome (b)]. Limitations of current speech technologies [Outcome (d)] will also be asked in the exam.	
Student Study Effort Expected	Class contact:	
	▪ Lectures and tutorials (8)	24 Hrs.
	▪ Laboratory/Presentation sessions (5)	15 Hrs.
	Other student study effort:	
	▪ Group project, presentation slides and poster	22 Hrs.
	▪ Writing laboratory report and self-learning	45 Hrs.
	Total student study effort	106 Hrs.
Reading List and References	[1] D. Jurafsky and J. H. Martin, <i>Speech and Language Processing</i>, 3rd Draft, 2023. [2] I. V. McLoughlin, <i>Speech and Audio Processing: A MATLAB®-Based Approach</i>. Cambridge: Cambridge University Press, 2016. [3] X. Tan, <i>Neural Text-to-Speech Synthesis</i>, in Artificial Intelligence: Foundations, Theory, and Algorithms series. Springer, 2023. [4] M.W. Mak and J.T. Chien, <i>Machine Learning for Speaker Recognition</i>, Cambridge University Press, 2020. [5] L. R. Rabiner and R. W. Schafer, <i>Theory and Applications of Digital Speech Processing</i>, Pearson, 2010. [6] X. Huang, A. Acero, H.-W. Hon, <i>Spoken Language Processing: A Guide to Theory, Algorithm, and System Development</i>, Prentice Hall, 2001. [7] J. R. Deller, J. G. Proakis, and J. H. L. Hansen, <i>Discrete-Time Processing of Speech Signals</i>, Macmillan Pub. Company, 2000. [8] D. Foster, <i>Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play</i>, 2nd Edition, O'Reilly, 2023. [9] A. Ng, <i>Introduction to Generative AI</i>, Coursera, 2023. [10] Simon J. D. Prince, <i>Understanding Deep Learning</i>, The MIT Press, 2023.	

August 2025